FIN 4000 Midterm Calculation Sheet

Formulas, variable definitions and worked examples for Chapters 2, 3, 4, 14, 15 and 16

FIN 4000 Investments: Midterm Calculation Sheet

This sheet collects every calculation used in the midterm and practice questions. Notation follows Bodie, Kane and Marcus (BKM). Each formula is followed by a definition of every symbol and a worked example. The examples use their own numbers, so they are not midterm or practice questions. Chapters 1 and 5 contain no calculations (they are conceptual questions), so they do not appear here.

Conventions used throughout

- Rates are written as decimals inside formulas (for example, 5% is 0.05) and as percentages in the worked examples.
- Unless a question says otherwise, interest compounds once a year, and coupon bonds have a face value (par value) of $1,000.
- Round only at the end of a calculation; intermediate rounding can change the last digit of your answer.

Calculator setup: TI BA II Plus

Do these once before the exam, and check them whenever a result looks odd.

Setting	Keys	Why
Show more decimals	`2ND FORMAT 9 ENTER 2ND QUIT`	Shows 9 decimal places so you can round only at the end
One payment per year	`2ND P/Y 1 ENTER 2ND QUIT`	Makes `I/Y` an annual rate and `N` a count of years. For semiannual problems, either keep $P/Y = 1$ and use $N = 2T$ with the per-period rate, or set $P/Y = 2$.
Clear the time value of money registers	`2ND CLR TVM`	Run before every new TVM problem
Payments at period end	`2ND BGN` shows `END` (if it shows `BGN`, press `2ND SET`, then `2ND QUIT`)	Bond coupons are paid at the end of each period
Clear the memories	`2ND MEM 2ND CLR WORK`	Memories `STO 1`–`STO 9` are used below to hold intermediate results

Reading the key notation. A key written in backticks is pressed once. `CPT` followed by a TVM key (`N`, `I/Y`, `PV`, `PMT`, `FV`) solves for that register. In the TVM registers, money you pay out is negative and money you receive is positive, so a bond's `PV` is entered with `+/-` and the other cash flows are positive. `I/Y` is entered as a percent (type `5` for 5%). `y^x` is the power key, and `1/x` is the reciprocal key. Results shown in the **Display** lines are rounded.

Chapter 2: Asset Classes and Financial Instruments

2.1 Tax-equivalent yield

Restate a tax-free municipal yield as the taxable yield that leaves the investor with the same amount after tax:

$$r_{TEY} = \frac{r_{muni}}{1 - t}$$

The after-tax yield of a taxable bond, and the tax rate at which the two bonds are equally attractive, are:

$$r_{AT} = r_{taxable}\,(1 - t) \qquad\qquad t^{*} = 1 - \frac{r_{muni}}{r_{taxable}}$$

Symbol	Definition
r_{TEY}	Tax-equivalent yield: the taxable yield that matches the municipal bond after tax
r_{muni}	Yield of the municipal bond (free of federal income tax), as a decimal
$r_{taxable}$	Yield of the taxable bond, as a decimal
r_{AT}	After-tax yield of the taxable bond
t	Investor's marginal federal income tax rate, as a decimal
t^{*}	Break-even tax rate: above it the municipal bond is better, below it the taxable bond is better

Example. A municipal bond yields 4.20% and the investor's tax rate is 22%. Then $r_{TEY} = 4.20\%/(1-0.22) = 5.38\%$. A taxable bond yielding 5.50% at a 30% tax rate leaves $r_{AT} = 5.50\% \times (1 - 0.30) = 3.85\%$ after tax. The break-even tax rate between the two bonds is $t^{*} = 1 - 4.20/5.50 = 23.64\%$.

TI BA II Plus. (Plain arithmetic; no TVM keys.)

Step	Keys	Display
Tax-equivalent yield	`4.20 ÷ ( 1 - 0.22 ) =`	5.3846
After-tax yield	`5.50 × ( 1 - 0.3 ) =`	3.8500
Break-even tax rate (as a percent)	`1 - 4.20 ÷ 5.50 = × 100 =`	23.6364

Chapter 3: How Securities Are Traded

3.1 Rate of return on a margin purchase

The investor puts up equity, borrows the rest, and measures the return on the equity:

$$E = m_0\, N\, P_0 \qquad L = N\, P_0 - E \qquad I = i\, L$$

$$r_{margin} = \frac{N\,(P_1 - P_0) + N\,D - I}{E}$$

Symbol	Definition
r_{margin}	Rate of return on the investor's own money for the holding period
N	Number of shares purchased
P_0	Purchase price per share
P_1	Selling price per share at the end of the period
D	Dividend per share received during the period
m_0	Initial margin requirement: the fraction of the purchase paid with the investor's own money
E	Equity: the investor's own money invested
L	Loan from the broker
i	Interest rate on the loan
I	Dollar interest paid on the loan for the period

Example. An investor buys \$300 shares at \$20.00 with 50% initial margin, borrowing at 7%. The stock pays a \$0.40 dividend and is sold at \$24.00 after one year. Cost = \$6,000, equity E = \$3,000, loan L = \$3,000, interest I = \$210. Price gain = \$1,200, dividends = \$120. Net profit = \$1,110, so $r_{margin} = 1,110/3,000 = 37.00\%$.

TI BA II Plus. (Plain arithmetic. The parentheses keys `(` and `)` are above the number pad.)

Step	Keys	Display
Equity E	`0.5 × 300 × 20 = STO 1`	3,000
Loan L	`300 × 20 − RCL 1 = STO 2`	3,000
Interest I	`RCL 2 × 0.07 = STO 3`	210
Net profit	`300 × ( 24 − 20 ) + 300 × 0.4 − RCL 3 =`	1,110
Return r_{margin} (as a percent)	`÷ RCL 1 = × 100 =`	37.00

3.2 Stock price that triggers a margin call

The loan does not change while the price falls, so only the equity shrinks. A margin call occurs when equity as a fraction of the stock's current value falls to the maintenance margin:

$$\frac{P\,N - L}{P\,N} = m \quad \Longrightarrow \quad P_{call} = \frac{L/N}{1 - m}$$

Symbol	Definition
P_{call}	Stock price at or below which the broker issues a margin call
P	Current stock price per share
N	Number of shares
L	Amount borrowed from the broker (constant while the price moves)
L/N	Loan per share, equal to $P_0\,(1 - m_0)$
m	Maintenance margin: the minimum equity as a fraction of the stock's market value
m_0	Initial margin requirement
P_0	Purchase price per share

Example. A stock is bought at \$60.00 with 55% initial margin and a 30% maintenance margin (ignore interest and dividends). The loan per share is $L/N = 60 \times (1 - 0.55) = \$27.00$, so $P_{call} = 27.00/(1 - 0.30) = \$38.57$.

TI BA II Plus.

Step	Keys	Display
Loan per share L/N	`60 × ( 1 - 0.55 ) =`	27.00
Margin call price P_{call}	`÷ ( 1 - 0.3 ) =`	38.57

The second step continues from the first display, so do not press `CE|C` in between.

3.3 Rate of return on a short sale

The short seller sells borrowed shares, buys them back later, owes any dividends to the lender, and posts margin as equity:

$$E = m_0\,N\,P_0 \qquad\qquad r_{short} = \frac{N\,(P_0 - P_1) - N\,D}{E}$$

The most a short seller can gain is $N\,P_0$ (the price cannot fall below zero). The loss has no upper limit because the price can rise without limit.

Symbol	Definition
r_{short}	Rate of return on the margin posted for the holding period
N	Number of shares sold short
P_0	Price at which the shares are sold short
P_1	Price at which the shares are bought back
D	Dividend per share paid to the lender while the position is open
m_0	Initial margin as a fraction of the value of the short position

Symbol	Definition
E	Equity posted as margin

Example. 300 shares are sold short at \$25.00 with 50% margin. During the year a \$0.30 dividend is paid, and the shares are bought back at \$20.00. Equity $E = \$3{,}750$. Gain on price = \$1,500, dividends owed = \$90, net profit = \$1,410, so $r_{short} = 1{,}410/3{,}750 = 37.60\%$.

TI BA II Plus.

Step	Keys	Display
Equity E	`0.5 × 300 × 25 = STO 1`	3,750
Net profit	`300 × ( 25 − 20 ) − 300 × 0.3 =`	1,410
Return r_{short} (as a percent)	`÷ RCL 1 = × 100 =`	37.60
Maximum possible gain	`300 × 25 =`	7,500

There is no key sequence for the maximum loss because it has no upper limit.

Chapter 4: Mutual Funds

4.1 Ending value of a fund investment after loads and expenses

A fund's annual expense ratio reduces the return every year. A front-end load is deducted once at purchase. A back-end load is deducted once at sale (and is often zero after a number of years). The questions use the approximation that net annual return equals gross return minus the expense ratio:

$$V_T = V_0 \,(1 - \ell)\,(1 + g - e)^{T}\,(1 - c_T)$$

Set $\ell = 0$ and $c_T = 0$ for a no-load fund. The fund with the larger ending value had the smaller total fee drag.

Symbol	Definition
V_T	Value of the investment after T years
V_0	Amount invested
ℓ	Front-end load, as a fraction of the amount invested (0 if none)
c_T	Back-end load charged at sale after T years, as a fraction of the value (0 if none or if the schedule has expired)
g	Annual gross return before expenses
e	Annual expense ratio (including any 12b-1 fee)
T	Number of years held

Example. \$10,000 is invested for 5 years with a gross return of 6%. A no-load fund with expenses of 1.50% ends at $V_T = 10{,}000 \times (1 + 0.06 - 0.015)^5 = \$12{,}461.82$. A fund with a 4% front-end load

and 0.80% expenses ends at $V_T = 10{,}000 \times (1 - 0.04) \times (1 + 0.06 - 0.008)^5 = \$12{,}369.44$. Over this horizon the front-end load fund has the larger fee drag.

TI BA II Plus (time value of money). Treat the net annual return $g - e$ as the interest rate. Use a negative `PV` for the amount that goes into the fund after any front-end load. Press `2ND CLR TVM` before each fund.

Step	Keys	Display
No-load fund	`5 N 4.5 I/Y -10000 PV 0 PMT CPT FV`	12,461.82
Front-end load fund: amount invested after load	`10000 × ( 1 - 0.04 ) =`	9,600.00
Front-end load fund: ending value	`2ND CLR TVM 5 N 5.2 I/Y +/- PV` (uses the display above) `0 PMT CPT FV`	12,369.44

For a back-end load, take `FV` and then multiply by $(1 - c_T)$: `× ( 1 - c ) =`.

Chapter 14: Bond Prices and Yields

14.1 Coupon interest

$$\text{Annual interest} = F \times c \qquad\qquad C = \frac{F \times c}{n}$$

Symbol	Definition
F	Face (par) value of the bond
c	Annual coupon rate, as a decimal
n	Number of coupon payments per year (1 annual, 2 semiannual, 4 quarterly)
C	Dollar amount of one coupon payment

Example. A bond has face value \$2,000 and a 5.5% coupon paid semiannually. Annual interest = $\$2{,}000 \times 0.055 = \$110$, and each payment is $C = \$55$.

TI BA II Plus.

Step	Keys	Display
Annual interest	`2000 × 0.055 =`	110.00
Payment per period	`÷ 2 =`	55.00

14.2 Free cash flow

$$FCF = CFO - CapEx$$

Symbol	Definition
FCF	Free cash flow: cash left after the investment needed to maintain and grow the business
CFO	Cash flow from operations
$CapEx$	Capital expenditures (spending on equipment, buildings and other long-lived assets)

Example. Cash from operations is \$8.0 billion and capital spending is \$5.5 billion, so $FCF = 8.0 - 5.5 = \$2.5$ billion. A negative result means investment exceeded the cash the operations produced.

TI BA II Plus.

Step	Keys	Display
Free cash flow	`8.0 - 5.5 =`	2.50

14.3 Day count and accrued interest

$$AI = C \times \frac{d}{D}$$

The convention decides how the days d and D are counted:

- **30/360** (U.S. corporate and municipal bonds): every month has 30 days and every year has 360 days.

$$d = 360\,(Y_2 - Y_1) + 30\,(M_2 - M_1) + (D_2 - D_1)$$

If a date falls on the 31st, treat it as the 30th. Under 30/360 a six-month coupon period always has $D = 180$. - **Actual/actual** (U.S. Treasury notes and bonds): d is the real number of days since the last coupon and D is the real number of days in the coupon period. - **Actual/360** (money market instruments): d is the real number of days elapsed and the year is taken as 360 days.

Symbol	Definition
AI	Accrued interest: the part of the next coupon already earned by the seller
C	Dollar amount of one coupon payment
d	Days from the last coupon date to the settlement date (counted by the convention)
D	Days in the full coupon period (counted by the convention)
Y_1, M_1, D_1	Year, month and day of the earlier date (last coupon)
Y_2, M_2, D_2	Year, month and day of the later date (settlement)

Example (30/360). A 6% semiannual bond with \$1,000 face has $C = \$30$. The last coupon was paid on February 10 and the trade settles on July 25 of the same year. Then $d = 30\,(7-2) + (25-10) = 165$, and $AI = 30 \times 165/180 = \$27.50$.

Example (actual/actual). The last coupon was paid on January 15, the trade settles on April 10, and the next coupon is due July 15 of the same year. Real days: $d = 85$ and $D = 181$, so $AI = 30 \times 85/181 = \$14.09$.

TI BA II Plus (date worksheet). The calculator counts days between two dates, and you choose the convention.

1. Open the worksheet: `2ND DATE`. Dates are typed as month.dayyear (month and day, then the two-digit year), so February 10, 2027 is `2.1027`.
2. `DT1` = first date, then `ENTER ↓`. `DT2` = second date, then `ENTER ↓`.
3. At `ACT/360`, press `2ND SET` to toggle between **ACT** (actual days) and **360** (30/360). Leave it on the one you need.
4. At `DBD` press `CPT` to get the number of days.
5. Multiply: coupon `×` days `÷` days in the period `=` to get AI.

Case	Entries	`DBD` shows	Then
30/360 days since the last coupon (d)	`DT1` = `2.1027`, `DT2` = `7.2527`, set to `360`, `CPT`	165	`30 × 165 ÷ 180 =` shows 27.50
Actual days since the last coupon (d)	`DT1` = `1.1527`, `DT2` = `4.1027`, set to `ACT`, `CPT`	85	(keep this number)
Actual days in the coupon period (D)	`DT1` = `1.1527`, `DT2` = `7.1527`, `CPT`	181	`30 × 85 ÷ 181 =` shows 14.09

Under 30/360 a semiannual coupon period is always 180 days, so you can type `180` directly.

14.4 Bond price from a yield

$$P = \sum_{t=1}^{T} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^T} = C\,\frac{1-(1+y)^{-T}}{y} + \frac{F}{(1+y)^T}$$

Symbol	Definition
P	Bond price (the present value of its cash flows)
C	Coupon payment per period
F	Face (par) value, paid at maturity
y	Yield to maturity per period, as a decimal
T	Number of periods until maturity
t	Period in which a cash flow is received

Example. A 4-year bond pays a 5% annual coupon ($50) on $1,000 face and the yield is 6%. $P = 50/1.06 + 50/1.06^2 + 50/1.06^3 + 1{,}050/1.06^4 = \$965.35$.

TI BA II Plus (time value of money). Solve for `PV` and ignore its sign.

Step	Keys	Display
Clear	`2ND CLR TVM`	
Enter the bond	`4 N 6 I/Y 50 PMT 1000 FV`	

Step	Keys	Display
Solve for price	`CPT PV`	-965.35 (the price is 965.35; the sign only shows it is a cash outflow)

For a semiannual bond, use $N = 2T$, $I/Y = y/2$ and the semiannual coupon, or set $P/Y = 2$ and enter N in years times 2.

14.5 Flat (clean) price and full (invoice) price

$$P_{full} = P_{flat} + AI$$

Symbol	Definition
P_{flat}	Quoted (clean) price, without accrued interest
P_{full}	Invoice (dirty) price the buyer actually pays
AI	Accrued interest from section 14.3

Example. If the quoted price is \$965.35 and accrued interest is \$27.50, the buyer pays \$992.85.

TI BA II Plus.

Step	Keys	Display
Full price	`965.35 + 27.50 =`	992.85

To find the flat price from a full price, subtract: full price `-` accrued interest `=`.

14.6 Yield to maturity (YTM)

The yield to maturity is the single discount rate y that makes the present value of the bond's promised cash flows equal its price:

$$P = \sum_{t=1}^{T} \frac{CF_t}{(1+y)^t}$$

It has no closed form for more than one period, so solve it with a calculator or by trial and error.

Symbol	Definition
P	Current price of the bond
CF_t	Cash flow in period t (coupon, plus face value in the last period)
y	Yield to maturity per period
T	Number of periods to maturity

TI BA II Plus (time value of money keys). Set one payment per year (annual coupons): `2ND P/Y 1 ENTER 2ND QUIT`. Clear with `2ND CLR TVM`. Enter `N` = number of periods, `PV` = price entered as a

negative number (+/-), PMT = coupon, FV = face value, then press CPT I/Y. For semiannual coupons use $N = 2T$, the semiannual coupon, and double the answer to quote an annual rate.

Example. A 3-year bond with a 7% annual coupon sells for \$1,024.00. Keys: N = 3, PV = −1,024.00, PMT = 70, FV = 1000, CPT I/Y gives YTM = 6.10%.

Compare with a yield you were given. To check a YTM answer, enter N, PMT, FV and the YTM in I/Y, then CPT PV. If the result equals the price (ignoring the sign), the YTM is right.

Chapter 15: The Term Structure of Interest Rates

15.1 Spot rates from a Treasury bill and a coupon bond (bootstrapping)

A 6-month Treasury bill pays only F at month 6, so its price gives the first spot rate directly. The one-year bond's first coupon is then discounted at that rate, and what remains of the price is the present value of the final payment:

$$1 + z_1 = \frac{F}{P_{bill}} \qquad y_1 = 2\,z_1$$

$$(1 + z_2)^2 = \frac{C + F}{P_{bond} - \dfrac{C}{1 + z_1}} \qquad y_2 = 2\,z_2$$

Symbol	Definition
P_{bill}	Price of the 6-month Treasury bill
P_{bond}	Price of the 1-year coupon bond
F	Face value (\$1,000)
C	Semiannual coupon payment of the bond
z_1	6-month spot rate for one 6-month period
z_2	12-month spot rate per 6-month period (the rate earned each half year over the full year)
$y_1,\ y_2$	Annual spot rates on a bond-equivalent basis (the 6-month rate times 2)

Example. A 6-month bill sells for \$965.00. A 1-year bond with a 7% coupon, paid semiannually (\$35 each), sells for \$1,010.00. Then $z_1 = 1000/965.00 - 1 = 3.627\%$ and $y_1 = 7.25\%$. The first coupon is worth $35/1.03627 = \$33.78$, leaving \$976.23. So $(1 + z_2)^2 = 1,035/976.23 = 1.06021$, $z_2 = 2.966\%$ and $y_2 = 5.93\%$.

TI BA II Plus. Use TVM for each spot rate. The answers are per 6-month period, so multiply by 2 for the bond-equivalent annual rate.

Step	Keys	Display
z_1: clear, then enter the bill	`2ND CLR TVM 1 N -965.00 PV` (type `965.00` then `+/-`) `0 PMT 1000 FV CPT I/Y`	3.6269
$y_1 = 2z_1$	`× 2 =`	7.2539
Present value of the first coupon	`35 ÷ ( 1 + 0.036269 ) =` (use the unrounded z_1: `RCL` it from memory if you stored it)	33.7750
What remains of the bond price	`1010 −` (the value above) `=`	976.2250
z_2: clear, then enter the final payment	`2ND CLR TVM 2 N 976.2250 +/- PV 0 PMT 1035 FV CPT I/Y`	2.9663
$y_2 = 2z_2$	`× 2 =`	5.9326

Tip: after the first `CPT I/Y`, press `STO 1` to keep z_1 (in percent). Later, `RCL 1 ÷ 100 + 1` gives $1 + z_1$.

15.2 Forward rates and discount factors

The forward rate for year n is the rate that, applied to year n, links the $(n-1)$-year and n-year spot rates:

$$(1+y_n)^n = (1+y_{n-1})^{n-1}\,(1+f_n) \quad\Longrightarrow\quad f_n = \frac{(1+y_n)^n}{(1+y_{n-1})^{n-1}} - 1$$

Discount factors are built by chaining the forward rates:

$$DF_1 = \frac{1}{1+f_1} \qquad DF_n = \frac{DF_{n-1}}{1+f_n} = \frac{1}{(1+y_n)^n}$$

Symbol	Definition
y_n	Spot rate (yield on an n-year zero-coupon bond), as an annual rate
f_n	One-year forward rate for year n (the rate for the period from year $n-1$ to year n); $f_1 = y_1$
DF_n	Discount factor: the present value of \$1 received at the end of year n
n	Number of years

Example. Spot rates are $y_1 = 4.0\%$, $y_2 = 4.8\%$ and $y_3 = 5.2\%$. Then $f_2 = 1.048^2/1.040 - 1 = 5.61\%$ and $f_3 = 1.052^3/1.048^2 - 1 = 6.00\%$. Discount factors: $DF_1 = 0.96154$, $DF_2 = 0.96154/(1+0.05606) = 0.91049$, $DF_3 = 0.91049/(1+0.06005) = 0.85892$.

TI BA II Plus. Spot rates are given in percent, so type them as decimals (for example, 4.8% is `0.048`).

Step	Keys	Display
f_2	`( 1.048 y^x 2 ) ÷ 1.04 - 1 =`	0.056062
f_3	`( 1.052 y^x 3 ) ÷ ( 1.048 y^x 2 ) - 1 =`	0.060046
DF_1	`1.04 1/x STO 1`	0.96154
DF_2 (chained)	`RCL 1 ÷ ( 1 + 0.056062 ) = STO 2`	0.91049
DF_3 (chained)	`RCL 2 ÷ ( 1 + 0.060046 ) = STO 3`	0.85892
Check DF_3 from the spot rate	`1.052 y^x 3 = 1/x`	0.85892

15.3 Bond price from discount factors, then its YTM

$$P = \sum_{t=1}^{T} CF_t \times DF_t$$

Then solve the yield to maturity from section 14.6 using that price.

Symbol	Definition
P	Bond price
CF_t	Cash flow in year t (coupon, plus face value in the last year)
DF_t	Discount factor for year t from section 15.2
T	Years to maturity

Example. Using the discount factors above, a 3-year bond with a 5% annual coupon ($50) has $P = 50 \times 0.96154 + 50 \times 0.91049 + 1050 \times 0.85892 = \$995.47$. Keys: `N` $= 3$, `PV` $= -995.47$, `PMT` $= 50$, `FV` $= 1000$, `CPT I/Y` gives YTM $= 5.17\%$.

TI BA II Plus. First the price from the stored discount factors, then the YTM from TVM.

Step	Keys	Display
Price	`50 × RCL 1 + 50 × RCL 2 + 1050 × RCL 3 = STO 4`	995.47
YTM: clear, then enter the bond	`2ND CLR TVM 3 N RCL 4 +/- PV 50 PMT 1000 FV CPT I/Y`	5.1669

If your memories were cleared, recompute the discount factors from section 15.2 first.

Chapter 16: Managing Bond Portfolios

16.1 Macaulay duration, modified duration and price change

$$w_t = \frac{CF_t/(1+y)^t}{P} \qquad D = \sum_{t=1}^{T} t\, w_t \qquad D^* = \frac{D}{1+y}$$

$$\frac{\Delta P}{P} \approx -D^{*}\,\Delta y$$

A zero-coupon bond's duration equals its maturity. The larger the duration, the more sensitive the price is to interest rate changes.

Symbol	Definition
w_t	Weight of period t: the present value of the cash flow in period t as a fraction of the price
CF_t	Cash flow in period t
y	Yield to maturity (annual)
P	Price (present value) of the bond or liability
D	Macaulay duration, in years
D^{*}	Modified duration
Δy	Change in yield, as a decimal
$\Delta P/P$	Percentage change in price

Example. A 3-year bond with a 6% annual coupon at a 5% yield has present values \$57.14, \$54.42 and \$915.67, so $P = \$1{,}027.23$. The weights are 0.0556, 0.0530 and 0.8914, giving $D = 2.8358$ years and $D^{*} = 2.8358/1.05 = 2.7007$. If the yield rises 0.5 percentage points, $\Delta P/P \approx -2.7007 \times 0.005 = -1.35\%$.

TI BA II Plus. Get each present value with TVM, store them, then combine. (A cash-flow worksheet gives the price directly: `CF 2ND CLR WORK`, `CF0 = 0`, `C01 = 60` with `F01 = 1`, `C02 = 60` with `F02 = 1`, `C03 = 1060` with `F03 = 1`, then `NPV`, `I = 5`, `↓`, `CPT`, which shows 1,027.23.)

Step	Keys	Display
PV of cash flow 1	`2ND CLR TVM 1 N 5 I/Y 0 PMT 60 FV CPT PV +/- STO 1`	57.1429
PV of cash flow 2	change `N` to `2` `60 FV CPT PV +/- STO 2`	54.4218
PV of cash flow 3	change `N` to `3` `1060 FV CPT PV +/- STO 3`	915.6679
Price P	`RCL 1 + RCL 2 + RCL 3 = STO 4`	1,027.2325
Macaulay duration D	`( 1 × RCL 1 + 2 × RCL 2 + 3 × RCL 3 ) ÷ RCL 4 = STO 5`	2.8358
Modified duration D^{*}	`RCL 5 ÷ ( 1 + 0.05 ) = STO 6`	2.7007
Price change for $\Delta y = +0.005$ (as a percent)	`RCL 6 × 0.005 = × 100 +/- =`	-1.3504

16.2 Convexity

Convexity corrects the duration estimate for large yield changes:

$$Cx = \frac{1}{P\,(1+y)^{2}} \sum_{t=1}^{T} \frac{CF_t}{(1+y)^{t}}\,(t^{2}+t) \qquad \frac{\Delta P}{P} \approx -D^{*}\,\Delta y + \tfrac{1}{2}\,Cx\,(\Delta y)^{2}$$

Symbol	Definition
Cx	Convexity of the bond
$CF_t,\ y,\ P,\ t$	As defined in section 16.1
D^*	Modified duration
Δy	Change in yield, as a decimal

Example. For the bond above, $Cx = 10.09$. For $\Delta y = +0.5$ percentage points, the duration-plus-convexity estimate is $\Delta P/P \approx -1.35\% + \frac{1}{2}(10.09)(0.005)^2 = -1.338\%$. The exact price change is -1.338%.

TI BA II Plus. This continues from section 16.1 and uses its stored present values (`STO` 1, 2, 3 and 4).

Step	Keys	Display
Sum of $PV_t \times (t^2 + t)$	`RCL 1 × 2 + RCL 2 × 6 + RCL 3 × 12 = STO 7`	11,428.8306
Convexity Cx	`RCL 7 ÷ ( RCL 4 × 1.1025 ) = STO 8`	10.0915
Convexity term $\frac{1}{2}\,Cx\,(\Delta y)^2$ (as a percent)	`0.5 × RCL 8 × 0.005 x² × 100 =`	0.0126
Duration plus convexity estimate (as a percent)	`RCL 6 × 0.005 × 100 +/- +` (the convexity term) `=`	-1.3377

Here $1.05^2 = 1.1025$ comes from $(1 + y)^2$. The multipliers 2, 6 and 12 are $t^2 + t$ for $t = 1, 2, 3$.

16.3 Immunizing a liability with two zero-coupon bonds

To protect a stream of payments against interest rate changes, choose a portfolio whose present value and duration equal those of the liability. For two zero-coupon bonds with maturities $T_1 < T_2$ (a zero's duration is its maturity):

$$PV_L = \sum_t \frac{L_t}{(1+y)^t} \qquad D_L = \sum_t t\,\frac{L_t/(1+y)^t}{PV_L}$$

$$w_1\,T_1 + (1 - w_1)\,T_2 = D_L \quad \Longrightarrow \quad w_1 = \frac{T_2 - D_L}{T_2 - T_1}, \qquad w_2 = 1 - w_1$$

$$\text{Dollars in bond } j = w_j \times PV_L \qquad \text{Bonds of \$1 face} = \text{Dollars} \times (1+y)^{T_j}$$

Symbol	Definition
L_t	Liability (payment due) in year t
y	Flat market yield, compounded annually
PV_L	Present value of the liability; the amount to invest today
D_L	Duration of the liability

Symbol	Definition
T_1, T_2	Maturities of the two zero-coupon bonds, in years
w_1, w_2	Fraction of PV_L invested in each zero-coupon bond
$(1+y)^{T_j}$	Growth factor that converts dollars invested into the face value that bond j pays at maturity

A negative weight means selling that bond short. Compare the liability's duration with a zero-coupon bond's maturity to answer sensitivity questions: if D_L is less than the bond's maturity, the liability is less sensitive to rate changes than that bond; if greater, it is more sensitive.

Example. Payments of \$5,000 are due in years 2 and 4 at a flat 6% yield. Present values are \$4449.98 and \$3960.47, so PV_L = \$8,410.45, and $D_L = 2.9418$ years. Using zeros maturing in 1 and 6 years: $w_1 = (6 - 2.9418)/(6 - 1) = 0.6116$ and $w_2 = 0.3884$. Dollars invested: \$5,144.17 and \$3,266.28. Bonds of \$1 face: $5,144.17 \times 1.06^1 = 5,453$ one-year zeros and $3,266.28 \times 1.06^6 = 4,633$ six-year zeros. Check: $w_1 T_1 + w_2 T_2 = 2.9418$, equal to D_L.

TI BA II Plus. Each step builds on the one before.

Step	Keys	Display
PV of the year 2 payment	`2ND CLR TVM 2 N 6 I/Y 0 PMT 5000 FV CPT PV +/- STO 1`	4449.9822
PV of the year 4 payment	change `N` to `4 CPT PV +/- STO 2`	3960.4683
Present value of the liability PV_L	`RCL 1 + RCL 2 = STO 3`	8,410.4505
Duration of the liability D_L	`( 2 × RCL 1 + 4 × RCL 2 ) ÷ RCL 3 = STO 4`	2.9418
Weight w_1	`( 6 - RCL 4 ) ÷ 5 = STO 5`	0.6116
Weight w_2	`1 - RCL 5 = STO 6`	0.3884
Dollars in bond 1	`RCL 5 × RCL 3 = STO 7`	5,144.17
Dollars in bond 2	`RCL 6 × RCL 3 = STO 8`	3,266.28
Number of \$1-face bonds, bond 1	`2ND CLR TVM 1 N 6 I/Y 0 PMT RCL 7 +/- PV CPT FV`	5,452.82
Number of \$1-face bonds, bond 2	change `N` to `6 RCL 8 +/- PV CPT FV`	4,633.28
Check the duration match	`RCL 5 × 1 + RCL 6 × 6 =`	2.9418

If w_1 comes out negative or above 1, one of the two bonds must be sold short. That is not an error.

16.4 Why immunization is difficult, and the simpler alternative: cash-flow matching

Difficulties with duration matching:

- **Rebalancing.** Duration changes as time passes and as rates change, and not at the same pace for the assets and the liability, so the portfolio must be rebalanced.
- **Limited protection.** It protects only against small, parallel shifts of the yield curve.
- **Convexity.** Differences in convexity leave the match imperfect for larger moves.

Cash-flow matching (dedication) buys a zero-coupon bond maturing on each payment date, with face value equal to the payment:

$$\text{Cost today} = \sum_t \frac{L_t}{(1+y)^t} = PV_L$$

Symbol	Definition
L_t	Payment due in year t (also the face value of the zero maturing in year t)
y	Yield, compounded annually
PV_L	Present value of the liability, which equals the cost of the matching zeros

Example. For the liability above, buy two zeros with \$5,000 face maturing in years 2 and 4. The cost is PV_L = \$8,410.45. Each bond pays the tuition exactly when it is due, so there is no rebalancing and no exposure to yield-curve shifts.

TI BA II Plus. The cost of the matching zeros is the same present value you found in 16.3. The year 2 and year 4 face values are the payments themselves, so no bond-count calculation is needed.

Step	Keys	Display
Cost of the year 2 zero	`2ND CLR TVM 2 N 6 I/Y 0 PMT 5000 FV CPT PV`	-4,449.98
Cost of the year 4 zero	change `N` to `4 CPT PV`	-3,960.47
Total cost	add the two present values (ignore the signs)	8,410.45

Quick reference: when to use which formula

Question type	Formula
Compare a municipal bond with a taxable bond	2.1
Return on a stock bought on margin	3.1
Price at which a margin call occurs	3.2
Return from selling a stock short; maximum loss or gain	3.3
Compare funds with different loads and expense ratios	4.1
Interest paid by a bond; free cash flow	14.1, 14.2
Accrued interest, day counts	14.3
Bond price from a yield; flat and full price; YTM	14.4, 14.5, 14.6
Spot rates from a bill and a coupon bond	15.1
Forward rates, discount factors, price from forwards, then YTM	15.2, 15.3, 14.6

Question type	Formula
Duration, price change, convexity	16.1, 16.2
Immunization and cash-flow matching	16.3, 16.4